

G H Raisoni College of Engineering and Management, Pune

(An Empowered Autonomous Institute affiliated to Savitribai Phule, Pune University, NAAC Accredited "A+" Grade)



T.Y B.Tech CSE (Data Science)

(TERM- I)

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23UDSPCL3508/23UDSPCP3508: Database Management Systems

1. Student Credentials

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- **Section:** T.Y B.Tech CSE (Data Science)

2. Database System Title

Title: Specialized Motorcycle Fleet Maintenance & Parts Inventory System

3. Problem Statement & System Requirements

Problem Statement:

Independent motorcycle garages and enthusiast collectives frequently rely on manual, physical ledgers to track long-term vehicle maintenance, mileage intervals, and specific high-performance lubricant inventories. This manual tracking leads to lost service histories, inconsistent oil change intervals, and inventory stockouts. This database aims to centralize service records, track odometer readings accurately, and manage the complex relationship between specific services and consumed parts.

Business Domain & Enterprise Entities:

- **Motorcycle:** The specific vehicle profile (e.g., Royal Enfield Bullet 350, 5-speed Cast Iron Electra).
- **Mechanic:** The technician performing the service.
- **Service_Record:** The transactional log of maintenance performed at a specific mileage.
- **Parts_Inventory:** The catalog of available spare parts and specific lubricants (e.g., Motul 300V, Motul 7100).

System Requirements & Business Rules:

- A **Motorcycle** requires multiple **Service_Records** over its lifespan (1:N).
- A **Service_Record** is performed by exactly one **Mechanic** (1:N).
- A **Service_Record** consumes many parts, and a specific part from the **Parts_Inventory** is used across many services (M:N relationship, requiring a junction table).
- Current service mileage must be strictly greater than the previous service mileage.

Primary Operational Transactions:

- **Insert:** Log a new completed service with the current odometer reading.
- **Update:** Deduct quantities from the inventory when parts are consumed.
- **Query:** Retrieve the complete service history and mileage tracking for a specific motorcycle chassis number.

4. Literature & System Reference

1. **"Automotive Fleet Maintenance Management Information Systems: A Relational Approach":** Analyzed best practices for decoupling static vehicle master data from continuous, time-series odometer tracking and service logs.
2. **ERP Inventory Module Case Studies:** Reviewed standard relational structures for resolving Many-to-Many (M:N) relationships in supply chains using "Line Item" junction tables to track parts consumed per work order.
3. **ISO 9001 Data Integrity Guidelines for Automotive Repair:** Referenced for enforcing strict domain constraints regarding sequential mileage logging and part traceability.

5. Conceptual Schema Design (ER / EER Diagram)

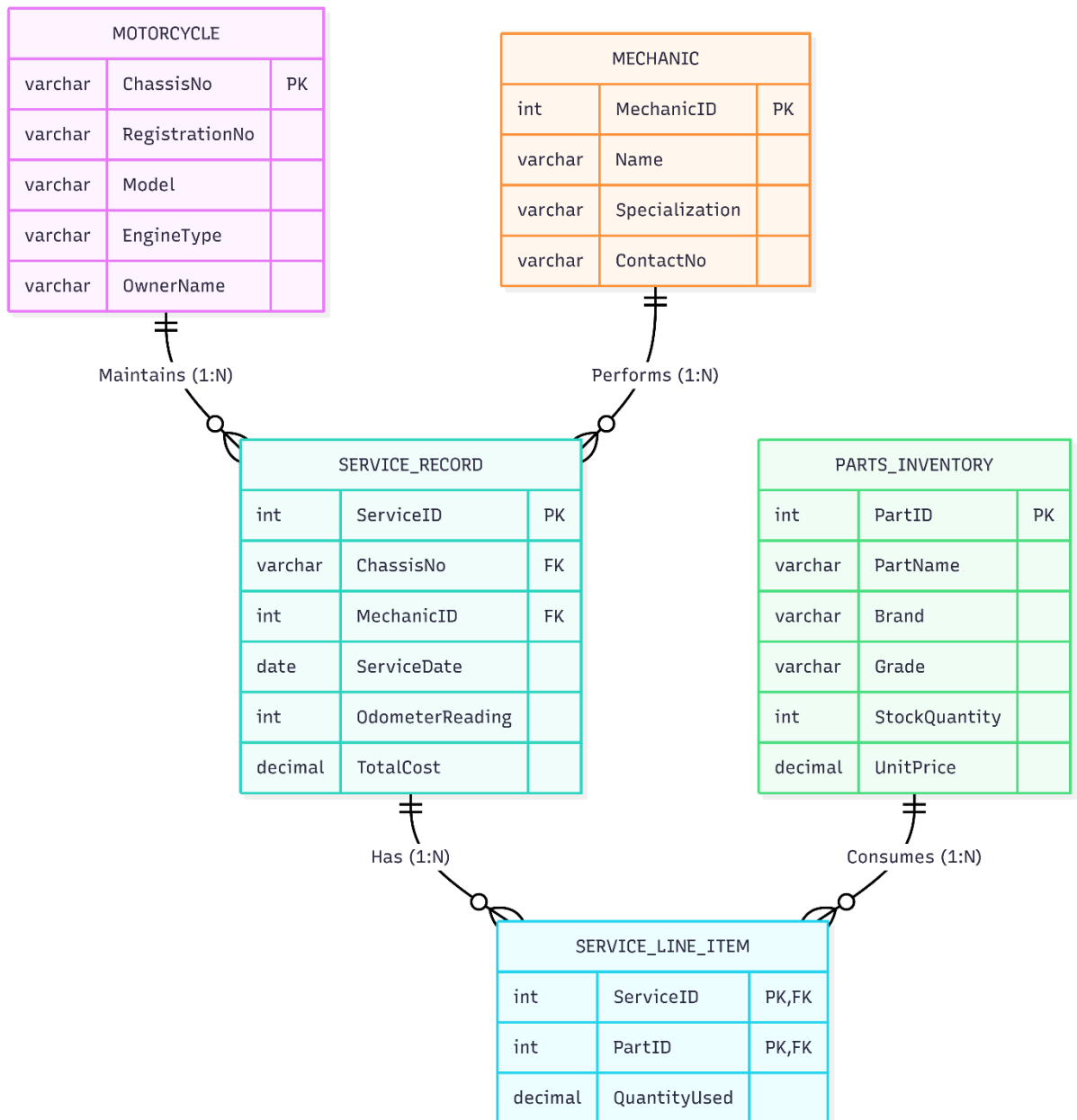
Entities, Attributes & Keys:

- **MOTORCYCLE:** ChassisNo (Primary Key), RegistrationNo, Model, EngineType, OwnerName.
- **MECHANIC:** MechanicID (Primary Key), Name, Specialization, ContactNo.
- **PARTS_INVENTORY:** PartID (Primary Key), PartName, Brand, Grade, StockQuantity, UnitPrice.
- **SERVICE_RECORD:** ServiceID (Primary Key), ChassisNo (Foreign Key), MechanicID (Foreign Key), ServiceDate, OdometerReading, TotalCost.

- **SERVICE_LINE_ITEM (Junction):** ServiceID (Foreign Key/Composite PK), PartID (Foreign Key/Composite PK), QuantityUsed.

Relationships & Cardinality Ratios:

- **Maintains (1:N):** MOTORCYCLE (1) connects to SERVICE_RECORD (N).
- **Performs (1:N):** MECHANIC (1) connects to SERVICE_RECORD (N).
- **Consumes (M:N):** SERVICE_RECORD (M) connects to PARTS_INVENTORY (N) via SERVICE_LINE_ITEM.



6. Relational Schema & Normalisation Plan

Step 1: Unnormalized Form (UNF)

Raw Data Source: A daily, handwritten operational log sheet used by garage staff. This data format is highly flawed. It suffers from repeating groups because multiple parts (and their quantities) are crammed into a single cell for one service event.

Chassis_No	Model	Odometer	Mech_ID	Mech_Name	Srvc_Date	Consumed_Parts_Log (Part, Qty)	Total_Cost
RE-ME123	Cast Iron Electra	43443	M01	Raj	2025-12-15	Motul 300V (2), Oil Filter (1), Spark Plug (2)	2800.00
RE-ME123	Cast Iron Electra	46101	M01	Raj	2026-04-20	Motul 300V (2), Chain Lube (1)	2700.00
RE-BU999	Bullet 350	12500	M02	Amit	2026-01-10	Motul 7100 (2.5), Brake Pad Front (1)	3325.00
RE-IN650	Interceptor 650	15200	M03	Vikram	2026-03-05	Motul 300V (3.2), Oil Filter (1), Spark Plug (4)	5670.00

Step 2: First Normal Form (1NF)

The Rule: A table is in 1NF if it contains no repeating groups, meaning every attribute (cell) must be atomic (single-valued). The Action: We must break apart the "Consumed_Parts_Log" column so that each part gets its own distinct row, duplicating the master service data for each part.

Chassis_No	Model	Odometer	Mech_ID	Mech_Name	Srvc_Date	Part_Name	Qty
RE-ME123	Cast Iron Electra	43443	M01	Raj	2025-12-15	Motul 300V	2.0
RE-ME123	Cast Iron Electra	43443	M01	Raj	2025-12-15	Oil Filter	1.0
RE-ME123	Cast Iron Electra	43443	M01	Raj	2025-12-15	Spark Plug	2.0

Step 3: Second Normal Form (2NF)

The Rule: A table is in 2NF if it is in 1NF and contains no partial dependencies. No non-prime attribute should depend on only a *portion* of the composite primary key. The Action: The Model attribute only depends on Chassis_No, not the Srvc_Date. The Mech_Name only depends on Mech_ID. To fix this, we logically separate the static master data from the transactional log.

- MOTORCYCLE: (Chassis_No, Model)
- MECHANIC: (Mech_ID, Mech_Name)
- SERVICE_LOG: (Chassis_No, Mech_ID, Srvc_Date, Odometer)
- PARTS_USED: (Chassis_No, Srvc_Date, Part_Name, Qty)

Step 4: Third Normal Form (3NF) & Boyce-Codd Normal Form (BCNF)

The Rule: A table is in 3NF if it has no transitive dependencies (where a non-key attribute dictates another non-key attribute). BCNF goes further, dictating that for every functional dependency ($X \rightarrow Y$), X must be a superkey. The Action: In our 2NF PARTS_USED table, relying on Chassis_No and Srvc_Date as a composite key is dangerous. If a bike is serviced twice in one day, the key fails. To achieve strict BCNF, we introduce independent surrogate primary keys (ServiceID and PartID). This ensures every determinant in our tables is definitively a primary key.

Final Relational Schema (BCNF Proof)

Below are the final, fully normalized tables populated with our dataset, proving the structural integrity of the design.

Table 1: MOTORCYCLE (Master Data)

ChassisNo (PK)	Model	EngineType	OwnerName
RE-ME123	Cast Iron Electra	350cc CI	Rushikesh S.
RE-BU999	Bullet 350	350cc UCE	Rohan K.
RE-IN650	Interceptor 650	648cc Parallel Twin	Sameer P.

Table 2: MECHANIC (Master Data)

MechanicID (PK)	Name	Specialization	ContactNo
M01	Raj	Royal Enfield CI & Modern	9876543210
M02	Amit	General Service	8765432109
M03	Vikram	Twin Cylinders & Superbikes	7654321098

Table 3: PARTS_INVENTORY (Master Data)

PartID (PK)	PartName	Brand	Grade/Spec	StockQuantity	UnitPrice
P01	Fully Synthetic Oil	Motul	300V 15W50	45	1100.00
P02	Synthetic Blend Oil	Motul	7100 20W50	60	850.00
P03	Oil Filter	RE Genuine	Standard	120	150.00
P04	Spark Plug	NGK	Iridium	80	450.00
P05	Chain Lube	Motul	C2	30	500.00

Table 4: SERVICE_RECORD (Transactional Data)

ServiceID (PK)	ChassisNo (FK)	MechanicID (FK)	ServiceDate	OdometerReading	TotalCost
S1001	RE-ME123	M01	2025-12-15	43443	2800.00
S1002	RE-BU999	M02	2026-01-10	12500	3325.00
S1003	RE-IN650	M03	2026-03-05	15200	5670.00
S1004	RE-ME123	M01	2026-04-20	46101	2700.00

Table 5: SERVICE_LINE_ITEM (Junction Table) (This table completely resolves the M:N relationship. Notice how Service S1001 uses multiple parts, and Part P01 is used across multiple services).

ServiceID (Composite PK, FK)	PartID (Composite PK, FK)	QuantityUsed
S1001	P01	2.0
S1001	P03	1.0
S1001	P04	2.0
S1002	P02	2.5
S1003	P01	3.2
S1003	P03	1.0
S1004	P01	2.0
S1004	P05	1.0

7. Target RDBMS Software

Target Software: MySQL 8.0 Justification: MySQL is chosen for Phase II of this project because of its incredibly robust handling of complex relational JOINS, which will be essential when querying data across the SERVICE_RECORD, SERVICE_LINE_ITEM, and PARTS_INVENTORY tables simultaneously.